

Edison Serrano

Applied Mathematician | ML Researcher | Scientific Computing

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🌐 Edison Serrano 📍 Bogotá, Colombia
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Professional Summary

Applied Mathematician and Data Scientist with research interests in Machine Learning, Artificial Intelligence, and Scientific Computing. Experienced in Bayesian inference, deep learning, optimization, and computational modeling, with applications in biomedical signal processing, natural language processing, and large-scale optimization. Passionate about developing mathematically grounded AI methods for complex real-world problems.

Education

MSc in Applied Mathematics

Center for Research in Mathematics (CIMAT)

Advisor: Marcos Capistrán

Relevant Coursework: Deep Learning, Inverse Problems, Data Science Techniques, Numerical Analysis and Optimization, Statistics, Analytical Modeling.

Aug 2023 - Sep 2025

Guanajuato, Gto., México

BSc in Mathematics

Universidad Nacional de Colombia (UNAL)

Advisor: Carlos Galvis

Relevant Coursework: Artificial Intelligence, Algorithms, Mathematical Modeling, Operating Systems, Theory of computation.

Aug 2018 - Oct 2023

Bogotá D. C., Colombia

Professional Experience

Research Assistant

University of Duisburg-Essen

Dec 2025 - May 2026

Essen, Germany

- **Developed and implemented algorithms for topology optimization** using topological derivatives and level set methods for compliance minimization in elastic structures.
- **Performed parallel simulations for shape and topology optimization** with FEniCSx within an AEI-DFG international research project.

Supervisor: Antoine Laurain

Research Assistant

Center for Research in Mathematics (CIMAT)

Nov 2023 - Dec 2023

Guanajuato, Mexico

- **Investigated numerical weather prediction models** by analyzing the Community Atmosphere Model (CAM), with emphasis on mathematical modeling and high-performance simulation

Supervisor: Silvia Jerez

Data Analyst

Empresa de Transporte del Tercer Milenio Transmilenio S.A

Dec 2022 - Jun 2023

Bogotá D. C., Colombia

- **Assessed data quality and developed monitoring dashboards** to automate brake pad wear alerts, monitor data transmission latency, and detect geolocation outliers across the bus fleet.
- **Built data processing pipelines** by integrating public data sources with cloud data warehouses, enabling automated monitoring of the new TransMilenio station door system and early detection of failures in bus components.

Supervisor: David Moreno

Publications

- **Hierarchical Attention Networks for Multilabel Sentiment Analysis in Spanish Reviews of Mexican Magic Towns**
IberLEF 2025, September 2025, Zaragoza, Spain
Authors: Torres Ezau, Edison Serrano
Proposes a lightweight hierarchical attention model for the joint classification of sentiment polarity, destination

type, and Magic Town from Spanish-language tourist reviews. Achieves strong results while being computationally efficient and interpretable, making it suitable for low-resource environments.

- **PACE Solver Description: CIMAT_Team**

International Symposium on Parameterized and Exact Computation (IPEC), 2024

DOI: [10.4230/LIPIcs.IPEC.2024.31](https://doi.org/10.4230/LIPIcs.IPEC.2024.31)

Authors: Carlos Segura, Lázaro Lugo, Gara Miranda, Edison David Serrano Cárdenas

Developed a memetic algorithm for the One-Sided Crossing Minimization (OCM) problem, which ranked first in the heuristic track of the Parameterized Algorithms and Computational Experiments Challenge 2024.

Invited Talks

- **Modeling a Diseased Heart: Simulation and Analysis of Cardiac Ischemia**

Feb 2025

Applied Mathematics Seminar, CIMAT Guanajuato, Mexico

- **Classification of Patients with Ischemia**

Dec 2024

Mini-Symposium on Applied Mathematics: Modeling, Scientific Computing, and Data Science

Annual Meeting of MexSIAM, CIMAT Mérida, Mexico

Remarkable Projects and Courses

- **Simulation and Analysis of Cardiac Ischemia** Sep 2025: Modeled ECG signals using dynamical systems and solved inverse problems for ischemia classification. Implemented Bayesian inference methods for parameter estimation and data-driven prediction.
- **Stein Variational Gradient Descent**, May 2025: Reviewed PyTorch implementations of Stein variational methods, analyzed algorithms, and conducted sampling simulations. [Github Repo](#)
- **Linux and Bash for Data Engineering Course**, Apr 2025: Covered Linux commands, shell scripting, process automation, file management, text processing, system permissions, and Bash-adjacent tools like Makefiles, Dockerfiles, and virtual environments. [Github Repo](#).
- **Advanced Deep Learning Course at CIMAT**, Jun 2024: Completed assignments on Extreme Learning Machines, Seq2Seq models, Transformer-Encoder architectures, LORA fine-tuning, and Denoising Diffusion Models for time series and image generation tasks. [Github Repo](#)
- **Research Assistant, Data Analyst** Jun 2022: Collaborated with a multidisciplinary team to analyze avocado leaf and sugarcane field data. Applied Convolutional Neural Networks for disease classification in avocado leaves and regression models to evaluate weather impacts on saccharose levels.
- **Software Development Project**, Agrosavia Jun 2022: Collaborated on a project with statistical geneticists and mathematicians to implement a R library to make selection signature analysis. [Github Repo](#)

Awards and Recognitions

International Mathematics and Programming Competitions: Developed advanced problem-solving, analytical thinking, teamwork, and time management skills while competing in prestigious mathematics and programming contests. Awarded first place in PACE Challenge 2024 and bronze medals in OMOUS 2022, IMC 2021, and CIIM 2021.

Teaching Experience

- **Teaching Assistant – Optimization Course**

Feb 2025 – Jun 2025

Universidad de Guanajuato, DEMAT

Assisted undergraduate students in the implementation of numerical optimization methods in python by providing academic support and grading assignments.

- **Teaching Assistant – Basic Mathematics**

Aug 2022 – Dec 2022

Universidad Nacional de Colombia (UNAL)

Assisted undergraduate students in foundational mathematics topics by providing academic support, designing and grading assignments.

Summer Schools and Workshops

- **Video Processing Course**
CIMAT Mérida, Mexico
Instructor: Dr. Francisco Javier Hernández López
Background subtraction, object detection, object tracking, and video analytics with applications in surveillance, autonomous vehicles, biomedical imaging, and industrial inspection.
 - **Computational Modeling and Simulation of Risks and Natural Disasters**
School of Modeling and Numerical Methods, CIMAT
Explored mathematical modeling and high-performance computing techniques for simulating floods, earthquakes, and epidemics.
 - **XXII School of Probability and Statistics**
University of Guadalajara, CUCEI, Mexico
Attended courses and lectures on bioinformatics, financial risk, infectious disease modeling, and machine learning.

Dec 2024

Jun 2024

Jun 2024

Technical Skills

- **Programming Languages:** Python, C++, SQL, R.
- **Machine Learning & Artificial Intelligence:** Scikit-learn, PyTorch, TensorFlow, XGBoost, NLTK, OpenCV.
- **Large Language Models & AI Agents:** OpenAI API, Hugging Face Transformers, LangGraph.
- **Scientific Computing & High-Performance Computing:** MPI, FreeFem++, FEniCSx, PyMC.
- **Data Engineering & Cloud:** PySpark, MongoDB, Google Cloud Platform (GCP), Looker Studio.
- **Software Development & MLOps:** Git, Docker, Linux, FastAPI, MLflow, SSH.

Languages

Spanish	Native
English	Proficient